Author's response to reviewer's comments
05-11-2026

Associate Editor and Reviewer comments:

AE, Dan Baldassarre: Thank you for your thorough revision and for bearing with us while we got reviews from the same two experts who reviewed your first draft. I'm happy to say that both reviewers were quite happy with your revised manuscript and the attention you paid to the issues they raised. At this point, there are mainly minor issues remaining, including:

1) Some areas where your conclusions/interpretations could be tempered a bit more.
2) More careful language around the nature of the RSF models and the "occupied/unoccupied dichotomy".
3) The need for another revision of the manuscript with an eye toward typos, formatting, inconsistent comparisons between the text and tables, etc.
We are thankful for the additional revisions and for all three points brought by the Associate Editor. We believe that those concerns were addressed in detail, as we have made sure to answer all the comments provided by the reviewers below.

Please see the detailed comments from the reviewers below and address all these points should you choose to submit a revision. I do not envision needing to send this back out for review if you thoroughly address these issues.

Reviewer #1: (The following can also be found in the attached doc)

Congratulations on a manuscript that is improved and, in my opinion, almost suitable for publication.
Most of my comments here relate to generally small fixes in the Methods, Results and Discussion that would improve the scientific quality and general clarity of the text. Please scan the document for any obvious errors like the ones I mention below.

Writing quality - While the writing quality is better than that in the previous submission, I have still found many instances in which the text lacks polish or could otherwise be much improved. Please pay close attention to the Introduction section as I did not provide comments on the writing quality there (for example, the second paragraph of the introduction needs a lot of work). I would like to see an improved version of this document in which the English is consistently at a level equivalent to that in similar publications in the field. Scan the document for typos.
The authors have revised the manuscript to scan for typos, grammatical errors or inconsistencies. We hope that this new version of the manuscript will appear more polished and refined compared to previous iterations.

Lines 122-123: Did you mean "...lower HUMAN population"? Make sure to add 'human' once in every section for extra clarity, unless you already used the word 'people' in the same context. Note that the first use of the word 'population' in the manuscript relates to cardinals, so you'll want to help the reader with context periodically.

We thank the reviewer for noting this important point. We indeed were discussing human population, and subsequent use of the term was adjusted to always make it clear when we are discussing the human population variable as opposed to "population" in another context.

Lines 170-176: Where's the transition? Please improve the text in these lines as it is confusing and potentially digressive.
- Note that in line 170 you end a sentence about what you did and start a new sentence about something that was done years ago and published in 2013. The text gives the reader the initial impression that you are still describing your own methods, or that you used 1.6 gr tags on your cardinals outside of the study during their reproductive season. You'll want to provide a fair transition and lay out your ideas in a more organized manner.
I suggest creating a new paragraph that starts with the premise of what you want to say before you provide the supporting evidence for it from Barron et al. Give the reader a chance to compartmentalize ideas.
We thank the reviewer for this insight. We have made the suggested changes and added a separate paragraph with a sentence to improve flow between the two sections. We hope that this brings organisation to this specific section of the manuscript.

The section called "Scale-of-effect": Consider moving this section after the Resource Selection Function section or adding some context. The reason is that you mention the predictor variables in this section without previously explicitly introducing them, which seems out of order.
We thank the reviewer for bringing this to our attention. After review, we have moved the "Scale-of-effect" section after the "Resource Selection Function" section.

Lines 249-287: Consider breaking up this large paragraph into two paragraphs. One option is to do so in line 266 to start a new paragraph with the sentence "We built our…".
The authors thank the reviewer for this suggestion, which was applied to the manuscript.

Figure 1:
- Maps require a north arrow as well as a scale bar.
- Mention the area, in hectares, of the sites. Do so in the caption and/or in the text.
We thank the reviewer for pointing out these details for Figure 1. We have adjusted the figure so that each site (A, B, and C) now has a scale bar and a north arrow. We also included the area covered for each site in hectares in the text, following the area in square kilometers.

Line 318 (and 448-450): Did you not include a statistical test for the difference in apparent survival rates that would allow you to say the rates differ statistically among sites? Unless I'm missing something, this seems necessary.
We thank the reviewer for highlighting this omission. To resolve this comment, we have included in the results section above Table 1, a detailed summary of the statistical analyses that were performed to evaluate differences in survival/tracking retention of the cardinals among sites. As a result, we have also updated the discussion section to reflect the new perspective that these analyses bring.

Line 339: Strictly speaking, given that you provided the normality test results before transformation, you need to provide them also after.

The authors thank the reviewer for pointing this out; we have reported the normality test results after the transformation to address this comment.

Table 2: I'm not convinced that Table 2 is necessary. I suggest rephrasing the paragraph starting in line 330 to include a sentence describing the core area for each site with mean +- SE, with the sample size for each in parentheses, followed by another sentence describing the home ranges in the same manner.
The authors thank the reviewer for this suggestion. In response, the paragraphs around Table 2 were significantly reworked, and Table 2 was removed from the manuscript.

Line 340: "although the effect approached significance" I'm in favor of leaving this sort of statement out, especially given that the next sentence describes the trend.
We agree with the reviewer and have removed this unnecessary sentence.
and
Lines 341-343:
- Rephrase to improve grammar.
- More importantly, there is an issue here with the statistical interpretation. You are trying to point out that the difference between one group and the others is important, but there is no evidence that this difference is statistically significant, i.e. more than just an artifact. This is, in my opinion, a little problematic. If you want to point to the sizes of the differences in log area then simply let the boxplot speak for itself and/or provide the mean with SE and the results of the multiple comparisons. Interpretation will come later, in the Discussion.
- "... a trend towards the cardinals at our urban site to have larger core home range sizes" implies an effect or an uncovered phenomenon, but the statistical evidence is insufficient for this sort of phrasing to be more than conjecture. Use objective wording; for example, describe which had the highest mean while noting the statistical non-significance among the groups.
The authors thank the reviewer for this detailed suggestion to improve the manuscript. We have worked on the section between lines 341-343 to remove subjective language and replace it by a paragraph that we hope reads more objectively. We have also revised the section to improve grammar and the overall language use throughout the section.

Line 344: We can use this as an example of writing quality that is not good enough. You mention "TukeyHSD" - Given that this technique is not mentioned in the methods, it has to at least be fully named here, as in "Tukey's Honestly Significant Difference". Simply mentioning the name of the R command is not appropriate.
We thank the reviewer for highlighting this specific example and have made the appropriate corrections.

Figure 4: Looking at the Urban sample size, why is the sample size 11 on the left (KDE 50%) and 9 on the right? Table 2 says the sample size for both is 9. Clarify this in the text and/or caption.
The authors appreciate this comment regarding the discrepancy between the two plots in Figure 4. We have revised the plot-generating code and have found a small error that caused the visual difference. We have revised it and have clarified this in the text, in conjunction with another comment suggesting that we remove Table 2 and adjust the paragraph accordingly.

Line 364, 366, 368 and Table S2: Did you mean "full" or "global" models? Be consistent, as these appear interchangeably.
We thank the reviewer for this comment. The terms were adjusted, and “global model” is used throughout.

Lines 365-366 and 369, 372: It looks to me like the reference to Table S1 belongs in the previous sub-section. The information here relates to Table S2. Also, referring to it multiple times seems redundant.
The authors thank the reviewer for this comment. We have indeed changed the reference to the appropriate table (S2 as opposed to previously S1), and have removed subsequent references to it after the first mention in the paragraph to improve the flow for the reader.

Paragraph 363-374: Use consistent terminology/format in reporting AIC scores.
I recommend sticking to the style:
$\Delta AICc < 10$
I think phrases like "< 2AICc units from…." and " > 10 D AICc values between…" are less preferable. Whatever you choose, be consistent.
The authors are grateful for the reviewer’s comment. To increase clarity and to improve consistency, we have rephrased our reporting of the AIC scores so that they are the same throughout the paragraph.

Lines 370-372 AIC difference: This is wrong according to the data in Table S2. The delta AIC is >3 given 444.7 vs. 447.9
We thank the reviewer for highlighting this. We have first made the appropriate change to refer to Table S2 instead of S1 in this section and have also adjusted the values according to the suggested change.

Paragraph 382 - 401:
- The opening sentence needs to be rephrased.
- Using words like "seem" is too colloquial and vague.
- Make sure it is clear which site/s you are referring to in every statement.
The authors thank the reviewers for their comment on this paragraph. The opening sentence was rephrased, and all mentions of “seem” throughout the manuscript were addressed. We also reviewed the paragraph to make sure the sites are clearly mentioned with every statement for clarity.

Figure 5 caption:
- The information in the caption should provide more detail to allow the figure to be independently understood. For example, you say "in a given radius area" - the radii need to be mentioned in the caption or figure.
- I believe that variables are in Table 3 in the current draft, not Table 2.
The authors thank the reviewer for these insights into the clarity of Figure 5. We have made additional changes to include more precision, as well as changed the reference to Table 3, which was indeed inaccurate in this draft of the manuscript.

DISCUSSION
Line 415: Please clarify in the text what UGS is before starting to use the initialism.
As recommended by the reviewer, we have included the full term "urban green spaces" before moving on to using the initialism UGS for the rest of the paragraph.

Line 422: Clarify that these are WINTER home ranges.
We agree with this comment by the reviewer and have changed "non-breeding season" to "winter season".

Table S1 - the Variables column is redundant, as the variables are the same in every row. Just describe the variables in the caption text.
The authors agree with this comment and have removed the "variables" column to include the different variables in the table caption instead.

Tables S2 - highlight the top model/s' AICc score in bold or some other way.
We agree with this suggestion by the reviewer and have made the top model's AICc score bold in Table S2.

Reviewer #2:
I appreciate the authors' careful attention to my previous comments. The manuscript is substantially improved. In particular, the translocation design is now much clearer, methodological limitations associated with tracking and spatial scale are more transparent, and the interpretations of home-range differences, apparent survival, and scale of habitat association have been appropriately tempered. I am satisfied that the major concerns raised in my initial review have been addressed, and I think the manuscript is now close to being suitable for publication.

I have only a few remaining minor comments.

1. RSF terminology: I have one remaining concern regarding the terminology used to describe the resource selection models. Table 3 describes the response as "presence/absence," while Figure 5 and the associated Results refer to "probability of presence." My understanding is that these models compare observed used locations with randomly generated available locations. Thus, the available locations are not observed absences. I suggest revising this terminology throughout to refer to used versus available locations and to describe the resulting relationships in terms of habitat use or habitat associations, rather than presence/absence or probability of presence.
We thank the reviewer for bringing this to our attention. We agree that adjusting the terms used to discuss habitat use in Figure 5, Table 3, and the rest of the manuscript would improve clarity and precision for the reader. Because our framework is not focused on occurrences, we have revised the specific sections mentioned in this comment, in addition to the rest of the manuscript, to remove mentions of "presence/absence" and replace them with "habitat use". We hope that these changes throughout the manuscript will improve the quality of the paper as a whole.

2. Behavioral plasticity: The Abstract concludes that "Cardinals exhibited behavioral plasticity in response to urbanization." I think this remains somewhat stronger than the data support. The

observed differences in space use and habitat associations across sites are consistent with behavioral flexibility, but the study does not directly quantify individual plasticity. I suggest tempering this statement accordingly.
The authors thank the reviewer for this comment. In response, we have changed the last sentence of the abstract, which was stronger than what the data support, to something more subtle, using the term "flexibility" instead. As a result, we have also changed the keywords to replace behavioral plasticity with behavioral flexibility. We hope that these changes will satisfy the reviewer.

3. Home-range estimates: I appreciate the additional information about the number of relocations used to estimate individual home ranges. I remain cautious about this result because KDEs were estimated from as few as five relocations, and the urban comparison is based on only nine individuals. I do not think an additional analysis is necessarily required at this stage, but I think the uncertainty associated with these estimates should be acknowledged more explicitly in the interpretation. In particular, the Discussion currently opens on line 422 by stating that individuals at the urban site had the largest total home ranges without acknowledging these limitations. A brief qualification here would be appropriate. I also suggest revisiting the statement in the Abstract that urban birds "had larger home ranges compared to birds wintering in the suburban and peri-urban sites." This statement appears stronger than the reported results support. In the Results (lines 339-347), core home-range size did not differ significantly among sites (P = 0.077), and although there was an overall site effect for total home range, the authors state that the only significant post-hoc comparison was urban vs. peri-urban (P = 0.03). I recommend revising the Abstract language accordingly so that it accurately reflects these results.
We thank the reviewer for this suggestion and for providing such detailed recommendations to improve the language used in relation to the strength of the results. We have changed both the abstract and the discussion to be more cautious and acknowledge the limitations related to the results surrounding the home ranges of the birds. We hope that these changes are improving the manuscript's accuracy in reporting the results.

4. Minor point - avoid contractions in the manuscript. For example, "couldn't" (line 423) and "didn't" (line 430) should be "could not" and "did not," respectively. I recommend checking the manuscript for any others as well.
We thank the reviewer for this comment. The authors have scanned the manuscript and have removed contractions throughout the text. A thorough review of typos and minor errors was also conducted.